Object-Oriented System Analysis Design

11th November 2024

Examination Paper

Answer ALL questions.

Clearly cross our surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Answer ALL questions

Marks

Question 1

a) Define the following terms in the context of Object-Oriented Analysis and Design:

i) Inheritance **2**

ii) Product backlog grooming in the context of Agile development. **2**

iii) MoSCoW based requirements specification. **2**

b) Multiple choice questions: Please select ONE (1) from the options provided.

i) When using Use-Case Points for estimation, which of the following can be used to assess technical complexity? **1**

A. Response time or throughput performance objectives
B. Lead analyst capability
C. Difficulty of programming language
D. All the above
E. None of the above

ii) In object-oriented design, which of the following is a behavioural diagram? **1**

A. Class diagram
B. State diagram
C. Deployment diagram
D. Package diagram

iii) In OO, a property of a class can be described by a **1**

A. Message
B. Instantiation
C. Method
D. Attribute
E. All of the above

iv) In OO, an accessor method **1**

A. Changes the state of an object's attributes
B. Returns a result without affecting the state of the object
C. Is the same as a mutator method
D. Is the same as a command

Total 10 Marks

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Marks

Question 2

Outline FIVE (5) merits **and** FIVE (5) demerits of the Waterfall model. **10**

Total 10 Marks

Question 3

a) List the FOUR (4) core values of the Agile Manifesto **and** explain how the Unified process (UP) satisfies the principles of the Agile Manifesto. **8**

b) Name TWO (2) of the TWELVE (12) principles of the Agile Manifesto. **2**

Total 10 Marks

Question 4

a) In a Health Care System, where patients can make several appointments over time, the gender of a patient can be recorded as *male*, *female, binary* or *undeclared*. **4**

Produce an implementation level class diagram showing the class Patient with at least THREE (3) attributes of different data type, ONE (1) of which must be gender as an enumeration. Show that the patient can *self-check-in* for an appointment.

b) In the class diagram in question 4a, above, explain with an example why patient height and patient weight are best placed in a separate class. **6**

Add the height and weight classes to the class diagram including the relationships to the Patient class. Show that height and weight can be used to check a patient's Body Mass Index (BMI).

Total 10 Marks

Question 5

Identify **and** explain FIVE (5) reasons why a software development project might fail. **10**

Total 10 Marks

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Marks

Question 6

a) Describe THREE (3) types of object-oriented partitioning used for testing. **6**

b) Describe TWO (2) types of object-oriented integration testing. **4**

Total 10 Marks

Question 7

a) Draw a state diagram for the following piece of code logic: **7**

```
Get value1
Multiply value1 by 6 giving value2
   If value2 > 30 Print value1, value2
            Close
   Else
            Return to start
```

b) Provide THREE (3) of the purposes of UML sequence diagrams. **3**

Total 10 Marks

Question 8

a) Identify FIVE (5) metrics for measuring inheritance. **5**

b) Briefly explain what is meant by polymorphism and state ONE (1) advantage of polymorphism. **2**

c) Provide ONE (1) polymorphism example based on a cooking app. **3**

Total 10 Marks

Question 9

a) Explain what is meant by a User Story in Agile. **4**

b) Provide a User Story for a Warehouse Clerk who is responsible for fulfilling orders in an Order Processing System. **6**

Total 10 Marks

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Marks

Question 10

Draw a UML Use-Case Diagram to illustrate the following process for an online purchase and delivery of products. Customers can make orders themselves or by phone. **10**

1. Customers or Orders Clerks select ONE (1) or more products from a product selection page.
2. Customers or Orders Clerks select checkout and enter payment details.
    a. If payment is not valid, customers are returned to the Checkout page.
    b. If payment is valid, customers select a delivery method (collection or shipping) and complete appropriate details.
3. The system confirms the order and updates the stock record.
4. The warehouse fulfils the order.

Total 10 Marks

End of paper

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